

Generation Interconnection System Impact Study Report

For Queue Project Q11-0018

Woodside 138 kV

April 28, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0018. The Interconnection Customer proposes a 1035.1 MW natural gas combined cycle facility interconnecting at the Woodside 138 kV substation in the Pennsylvania area. The request was received on March 08, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 13 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$113,347,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Woodside Substation - Milesburg Substation 138.0 kV, the Sprintbrook - Woodside 138.0 kV Line loads to 473.1% of its 215 MVA rating. Project DFAX: 100.0%.

The Milesburg Substation - Woodside 138.0 kV is overloaded to 394.7% of its 258 MVA emergency rating for the loss of the Sprintbrook - Woodside 138.0 kV. The project contributes 100.0% DFAX (1035.1 MW) to this overload.

Milesburg - Scott 138.0 kV: 368.1% loading (272 MVA) following the outage of the Sprintbrook - Woodside 138.0 kV; DFAX = 100.0%.

For the loss of Woodside - Milesburg 138.0 kV, the Sprintbrook - Orchard Road 138.0 kV loads to 246.1% of its 284 MVA rating. Project DFAX: 74.2%.

Orchard Rd 345/138 kV Autotransformer: 202.8% loading (344 MVA) following the outage of the Milesburg - Woodside 138.0 kV; DFAX = 74.2%.

The George St - Volcano 138.0 kV line is overloaded to 194.1% of its 165 MVA emergency rating for the loss of the Sprintbrook - Orchard Rd 138.0 kV Line. The project contributes 31.6% DFAX (327.3 MW) to this overload.

For the loss of Sprintbrook - Woodside 138.0 kV, the Scott 345/138kV Autotransformer loads to 169.7% of its 388 MVA rating. Project DFAX: 72.5%.

Lo Reales - El Cerrito Sta G 138.0 kV: 163.7% loading (183 MVA) following the outage of the Sprintbrook - Woodside 138.0 kV Line; DFAX = 27.5%.

The Los Reales Substation - Dougoil Substation 138.0 kV Line is overloaded to 150.0% of its 203 MVA emergency rating for the loss of the Sprintbrook - Woodside 138.0 kV Line. The project contributes 27.5% DFAX (284.6 MW) to this overload.

For the loss of Sprintbrook - Orchard Rd Substation 138.0 kV, the Charity Church - Orcas 138.0 kV loads to 141.0% of its 238 MVA rating. Project DFAX: 31.6%.

The Boc Gasses - Orcas 138.0 kV Line is overloaded to 135.9% of its 252 MVA emergency rating for the loss of the Sprintbrook - Orchard Rd 138.0 kV. The project contributes 31.6% DFAX (327.3 MW) to this overload.

For the loss of Sprintbrook - Woodside 138.0 kV, the Glory Rd - Dougol 138.0 kV loads to 135.3% of its 232 MVA rating. Project DFAX: 27.5%.

Boc Gasses - Sprintbrook 138.0 kV: 134.1% loading (260 MVA) following the outage of the Orchard Road - Sprintbrook 138.0 kV; DFAX = 31.6%.

Glory Rd - Indian River Substation 138.0 kV: 126.8% loading (259 MVA) following the outage of the Sprintbrook - Woodside Substation 138.0 kV; DFAX = 27.5%.

The Indian Riv - Scott Substation 138.0 kV Line is overloaded to 121.9% of its 282 MVA emergency rating for the loss of the Sprintbrook - Woodside Substation 138.0 kV L. The project contributes 27.5% DFAX (284.6 MW) to this overload.

For the loss of Orchard Road - Sprintbrook Substation 138.0 kV line, the Walnut Crk - Volcano 138.0 kV line loads to 107.4% of its 195 MVA rating. Project DFAX: 21.8%.

The Dekalb - Robison Park 230.0 kV is overloaded to 103.1% of its 777 MVA emergency rating for the loss of the Jones Chapel - Dekalb 500.0 kV. The project contributes 6.6% DFAX (68.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sprintbrook - Woodside 138.0 kV	Rebuild line with higher-capacity conductor	\$3,801,000	100% (\$3,801,000)
Milesburg Substation - Woodside 138.0 kV	Rebuild line with higher-capacity conductor	\$6,637,000	100% (\$6,637,000)
Scott Substation - Milesburg 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,495,000	100% (\$1,495,000)
Sprintbrook - Orchard Road 138.0 kV	Rebuild line with higher-capacity conductor	\$2,105,000	100% (\$2,105,000)
Orchard Rd 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$16,935,000	100% (\$16,935,000)
George St - Volcano 138.0 kV	Rebuild line with higher-capacity conductor	\$13,507,000	100% (\$13,507,000)
Scott Substation 345/138 kV Auto	Replace transformer with higher-rated unit	\$18,671,000	100% (\$18,671,000)
Lo Reales - El Cerrito Sta G 138.0 kV	Rebuild line with higher-capacity conductor	\$3,912,000	100% (\$3,912,000)
Los Reales Substation - Dougoil Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$8,144,000	100% (\$8,144,000)
Orcas - Charity Church 138.0 kV	Rebuild line with higher-capacity conductor	\$2,176,000	100% (\$2,176,000)
Boc Gasses - Orcas 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,405,000	100% (\$4,405,000)
Glory Rd - Dougoil 138.0 kV	Rebuild line with higher-capacity conductor	\$4,474,000	100% (\$4,474,000)
Boc Gasses - Sprintbrook 138.0 kV	Rebuild line with higher-capacity conductor	\$5,780,000	100% (\$5,780,000)
Glory Rd - Indian River 138.0 kV	Rebuild line with higher-capacity conductor	\$5,174,000	100% (\$5,174,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Indian Riv - Scott Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,599,000	100% (\$5,599,000)
Walnut Crk - Volcano 138.0 kV	Reconductor line	\$1,309,000	100% (\$1,309,000)
Robison Park - Dekalb 230.0 kV	Reconductor line	\$10,017,000	92% (\$9,223,000)
	Total allocated to project		\$113,347,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.